



VST3 / AU PLUGIN · V0.13.0

Gabci's SmartMaskNetwork

Real-time, cross-track psychoacoustic spectral masking — instead of traditional one-way sidechain ducking, every loaded instance in the session continuously and mutually “talks” to every other one, so frequency clashes get resolved automatically.

Engine

C++20 / JUCE 9.0.0

Formats

VST3 · AU · Standalone

CPU / instance

~1%, measured live

Version

v0.13.0 · 2026

Not sidechain. A network.

Studio-standard sidechain ducking is one-directional: track A pushes track B down, but B never knows what A is doing. SmartMask Network wires every loaded instance into a single, lock-free shared registry instead — every track continuously sees every other track's real, psychoacoustic (24-band Bark-scale) spectrum, and automatically, surgically makes room for the more important sounds based on its own priority, instead of ducking the whole track or a fixed frequency range. The result: less manual EQ carving, a cleaner mix, and live visual feedback that shows you exactly when, where, and how much masking is actually happening.

Standout features

1

Two-way, priority-based masking

There's no fixed “master” track pushing everything else down — any track can mask any other, purely based on the configured priority order. That order can be edited from any single open instance by dragging; dropping onto an occupied slot automatically swaps the two tracks.

2

Live “collision” visualization

The spectrum view has a real, labelled Hz axis (20Hz–20kHz), dB gridlines, and a mouse-hover readout showing exactly which track dominates which frequency — and for your own track, a translucent fill shows precisely where and how much the system is masking it right now.

3

Click-free Bypass

Bypass doesn't tear out the whole processing chain — it just forces the mask gains to unity, so latency stays constant even while playing back, meaning you can A/B the effect without a single click or glitch.

4

Phase-safe stereo analysis

No time-domain L+R downmix before analysis — every channel gets its own FFT, and only the resulting energies are summed. A wide, partially out-of-phase stereo pad never “disappears” from the masking engine's view.

Low resource usage, industrial-grade internals



The plugin's internal architecture doesn't just “work” — it was built to the strictest rules of real-time audio processing, and every critical decision along the way was verified with an actual measurement, not a guess.

Lock-free, double-buffered registry. Cross-track communication happens exclusively through atomic pointer swaps — the audio thread never waits on another thread, so the plugin cannot, by design, cause an audio dropout even in a crowded session.

Zero allocation, verified. An automated test that overrides the global new/delete operators guarantees that not a single memory allocation happens inside the audio processing loop — every buffer is pre-allocated up front.

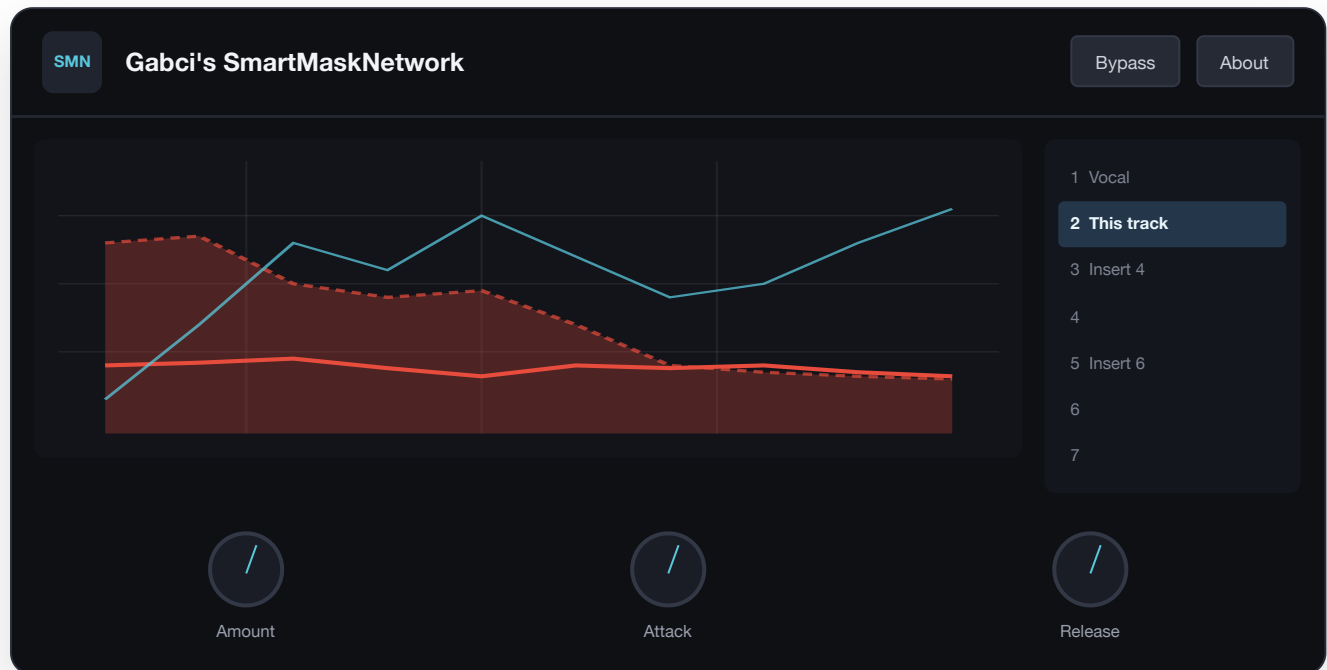
A cross-instance “request channel”, not a direct write. When one track changes another track's priority, it never reaches directly into that other instance's own parameter state — instead it posts a request, which the owning instance applies to itself. This matters because it means the host's own Undo/Redo and automation system is never broken for any track involved.

Measured, not textbook, latency. The plugin reports the real, measured 3584-sample processing delay to the host (not a theoretical estimate), so the DAW can always compensate for it exactly.

SIMD-accelerated signal processing. The critical gain-multiplication and overlap-add summation steps run as vectorized (SIMD-dispatched) operations, which measurably reduced the per-instance CPU load.

The interface, at a glance

The interface has three parts: the header, the spectrum visualizer in the middle with the priority list beside it, and the three adjustable parameters below.



- 1 Bypass**
Turns off masking's effect on this track with one click, with zero clicks or glitches — ideal for A/B comparison.
- 2 Spectrum visualizer**
Every active track's curve is shown; on your own track, the red fill shows exactly where and how much the system is masking it right now.
- 3 Priority list**
Drag any track to reposition it, even from a different instance's own window — dropping on an occupied slot swaps the two.
- 4 Amount / Attack / Release**
Amount is the maximum attenuation ceiling (0–100%); Attack/Release control how fast the system reacts to a level change in that track.

References & acknowledgements

- **JUCE 9.0.0** — the C++ audio application framework (juce.com) underlying the processing engine, the VST3/AU/Standalone packaging, and the user interface.
- **Steinberg VST3 SDK** and **Apple Audio Unit SDK** — the plugin format standards, including the optional track-name-reporting (IInfoListener channel context) mechanism.
- **Catch2** — the C++ unit-testing framework used for the full automated test suite (registry, DSP chain, real-time allocation guard).
- **E. Zwicker's psychoacoustic Bark-scale formula** (E. Zwicker, “Subdivision of the audible frequency range into critical bands”, Journal of the Acoustical Society of America, 1961) — the mathematical basis for the 24-critical-band frequency analysis.
- **ITU-R BS.1770** loudness-measurement principle — the rationale behind phase-safe, per-channel energy summation instead of a time-domain downmix.

This plugin was built by Claude Code (Sonnet 5), based on a joint idea by Gábor Tomori and Gemini, under Gábor Tomori's direction and with Gemini's review, in 2026.

Tested on Intel Mac OS 15.7, FL Studio 2026